1. Video Evaluation Framework Solution Approach

To address the challenge of assessing videos qualitatively and quantitatively, I developed a **modular multi-agent framework** using **LangGraph** to orchestrate the pipeline. The evaluation process is divided into three distinct stages:

1. **Perception Analysis**:
   I extract frames from the video and compute optical flow between them to assess **temporal coherence** — how smooth and stable the motion is. This is crucial for detecting jitter or unnatural transitions in generated content.
2. **Semantic Reasoning**:
   Using the **CLIP model**, I compare each frame to a user-defined text prompt (e.g., “a person using a smartphone”). This gives a **semantic similarity score**, helping me measure whether the video stays on-topic. I also detect **scene transitions** using pixel-wise differences to find abrupt cuts.
3. **LLM-Based Reporting**:
   Finally, I feed the computed metrics into a **Large Language model** (e.g., ChatGPT or Claude) to generate a structured technical report. This makes the output easy to understand, even for non-technical stakeholders.

All components are orchestrated using **LangGraph**, with agents communicating via **Shared Memory**. The evaluation is **criteria-driven**, configured through a simple criteria.json file — making the framework highly adaptable to various video types like ads, explainers, or short films.

2. System Architecture

To structure the evaluation process effectively, I designed the system using a **modular agent-based architecture** orchestrated by **LangGraph**. The overall architecture is broken into three sequential agents — each responsible for one aspect of the evaluation pipeline. The flow is as follows:

Agent Workflow (Orchestrated by LangGraph)

Video Path + Prompt + Evaluation Criteria (JSON)
↓
[Perception Agent] → computes frame data + optical flow
↓
[Reasoning Agent] → evaluates semantic alignment + scene transitions
↓
[Reporting Agent] → generates final report using an LLM

Each agent operates independently and contributes to the final evaluation by writing to a **shared memory dictionary** (implemented using Python’s multiprocessing.Manager().dict()), which ensures loose coupling while maintaining data continuity.

Shared Memory + JSON-Driven Configuration

Rather than hard-coding evaluation logic, I designed the system to accept a **criteria.json** input file. This makes the system **highly adaptable**. By using shared memory for agent communication and JSON for control, I built the architecture to be both **extensible and production-ready**.

3. Metric Definitions and Computation

Each agent in my system contributes quantitative metrics that reflect a specific aspect of video quality. Below is a breakdown of the metrics used in each agent and how I compute and interpret them.

3.1 Temporal Coherence (Perception Agent):

Definition:
Temporal coherence refers to the **smoothness of motion** between video frames. For ad-quality videos, abrupt or jittery movements can break immersion and reduce realism.
How I Compute It:

- I extract frames at 1 FPS using OpenCV.
- Then, I use the **Farneback method** in OpenCV to compute **dense optical flow** between consecutive frames.
- From the optical flow vectors, I compute the **magnitude of motion** between frames.
- Finally, I calculate:
    - **Average Optical Flow** – overall motion magnitude
    - **Standard Deviation** – jitter or inconsistency in motion

3.2 Semantic Consistency (Reasoning Agent):

Definition:
Semantic consistency measures how well the video content aligns with the intended theme or prompt — for example, whether the ad actually depicts a person using a smartphone.

How I Compute It:

- I use the **CLIP model** to generate embeddings for each frame and for the user-defined text prompt.
- I compute the **cosine similarity** between each frame and the prompt.
- I then calculate:
    - **Mean CLIP Score** – average alignment across all frames
    - **Clip Score Distribution** – used to visualize variance

3.3 Scene Transitions (Reasoning Agent):

Definition:
Detects whether the video has **abrupt or inconsistent scene changes**, which can harm narrative flow.
How I Compute It:

- I convert each frame to grayscale.
- Then I compute the **pixel-wise absolute difference** between consecutive frames.
- If the difference crosses a fixed threshold (e.g., 0.2 normalized), I mark it as a **scene transition**.

3.4 Final Report Summary (Reporting Agent):

Definition:
An LLM-generated technical summary based on all the above metrics, explaining:

- The quality of motion
- The semantic focus of the video
- Visual consistency and story flow
- A final **score out of 10**

How I Generate It:

- I pass a **formatted summary** of the perception and reasoning metrics to an LLM (e.g., ChatGPT or Claude).
- The LLM uses an instruction prompt to generate a natural language report.

Why It Matters:

- This layer brings **interpretability** to the metrics.
- It bridges the gap between raw numbers and actionable insights — critical for non-technical users, marketers, or creative teams.

4. Experimentation & Results

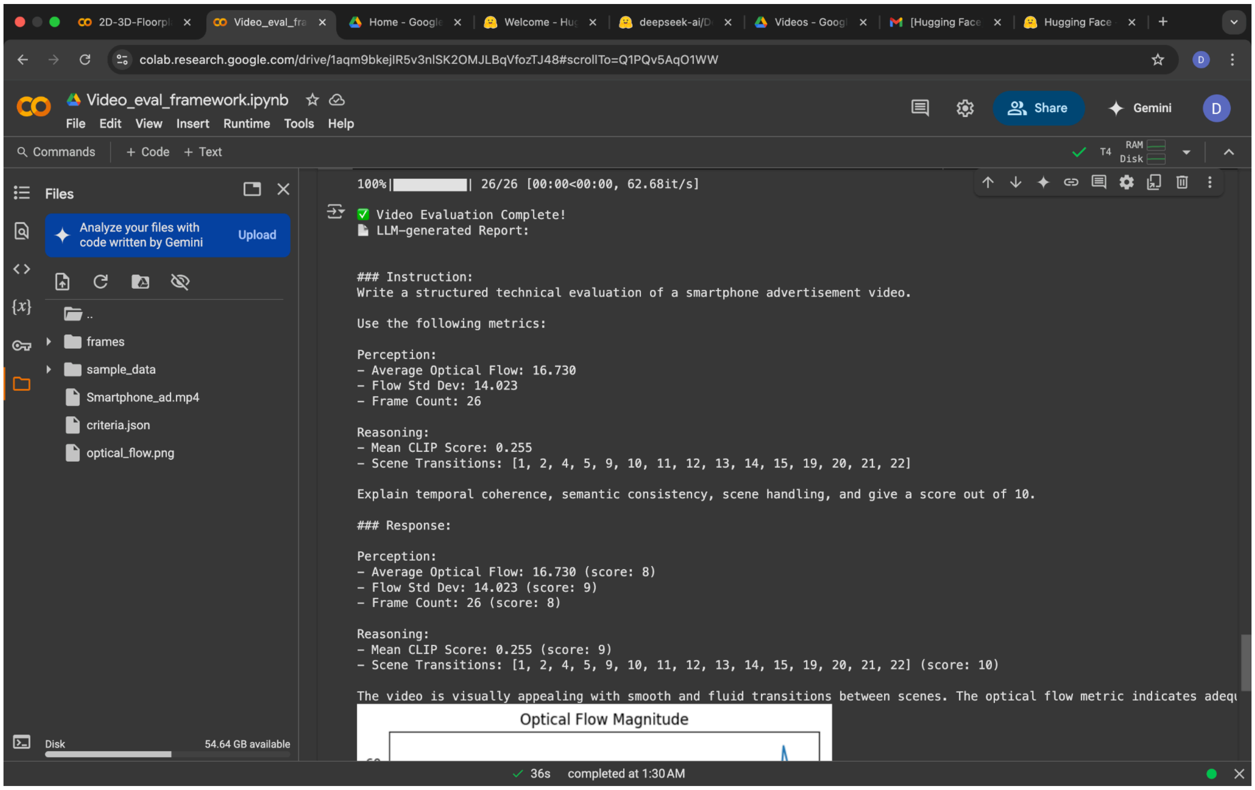
After implementing the complete video evaluation framework, **I tested it on the same AI generated Smartphone_ad video which I had submitted in the 1st assignment given.** The objective was to determine whether the system could produce interpretable, reliable insights about the video's quality — both from a **technical** and **semantic** standpoint.

Test Setup

- **Input Video**: AI generated Smartphone_ad video submitted in assignment 1.
- **Prompt Given to CLIP**: "`a premium smartphone being used by a person in everyday life situations`"
- **Model Used**: TinyLlama/TinyLlama-1.1B-Chat-v1.0 for testing purposes since I don't have access to OpenAI or Anthropic LLMs.
- **Evaluation Criteria**:
  As defined in criteria.json, I enabled:
  {
    "perception": { "temporal_coherence": true },
    "reasoning": { "semantic_consistency": true, "scene_transitions": true },

```
  "reporting": { "generate_summary": true }
}
```

Observed Outputs



Full response from Reporting Agent (LLM):

```
### Response:
Perception:
- Average Optical Flow: 16.730 (score: 8)
- Flow Std Dev: 14.023 (score: 9)
- Frame Count: 26 (score: 8)
Reasoning:
- Mean CLIP Score: 0.255 (score: 9)
- Scene Transitions: [1, 2, 4, 5, 9, 10, 11, 12, 13, 14, 15, 19, 20, 21,
22] (score: 10)

The video is visually appealing with smooth and fluid transitions between
scenes. The optical flow metric indicates adequate temporal coherence, but
the flow stdev is slightly high, indicating that the motion of objects in
the video is not consistently estimated. Additionally, the scene
transitions are aesthetically pleasing and help create a coherent
narrative. The score out of 10 is 8, indicating a moderate level of
perceived quality.
```